Experiment 3: Error Handling, Static Resources and Rate-Limiting Mechanisms

Aim:
To implement a simple web application using Express.js to demonstrate error handling, static resources and rate-limiting mechanisms.

Procedure

1. Create project folder
mkdir security-demo
cd security-demo

2. Initialize project
npm init -y

3. Install packages
npm install express express-rate-limit

4. Create project structure

```
security-demo/
├── public/
│   ├── index.html
│   ├── image.jpg
│   └── style.css
├── app.js
├── package.json
└── node_modules/
```

5. Create index.html

```
<!DOCTYPE html>
<html>
<head>
<title>Express Static Resource Demo</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<h1>Welcome to Express.js</h1>
<p>This page demonstrates static resources.</p>
```

```
<img src="image.jpg" width="300">
</body>
</html>
```

6. Create style.css

```
body{
 font-family: Arial;
 text-align: center;
}

h1{
 color: blue;
}

p{
 color: green;
}
```

7. Create app.js

```
const express = require('express');
const rateLimit = require('express-rate-limit');

const app = express();

app.use(express.static('public'));

const limiter = rateLimit({
 windowMs: 1 * 60 * 1000,
 max: 5,
 message: 'Too many requests, please try again later.'
});

app.use(limiter);

app.get('/hello', (req,res)=>{
 res.send('Hello Express.js');
});

app.use((req,res)=>{
 res.status(404).send('Page Not Found');
});
```

```
app.listen(3000, ()=>{
 console.log('Server running at http://localhost:3000');
});
```

8. Execute

```
node app.js
```

Outputs

http://localhost:3000/
Displays index.html with CSS and image.

http://localhost:3000/hello
Displays: Hello Express.js

Invalid URL
Displays: Page Not Found

After 5 requests in one minute
Displays: Too many requests, please try again later.

Result
Successfully demonstrated static resources, error handling and rate limiting in Express.js.